

JIMMA UNIVERSITY
JIMMA INSTITUTE OF TECHNOLOGY
FACULTY OF COMPUTING AND INFORMATICS



m
COMMUNITY BASED TRAINING PROGRAMME
INTERVENTION REPORT IN BOSA ADDIS KEBELE
REPORT ON PHASE - 3

By:
Computer Science Section 2, Group 5 Students

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Finally, we thank all group members for their commitment, teamwork, and contribution during development, testing, demonstration, user support, and documentation of the Phase Three intervention.

ABSTRACT

This Phase Three report presents the intervention and implementation activities conducted by Computer Science Section 2, Group 5 students in Bosa Addis Kebele. In Phase One, the group identified community problems related to limited technology use, manual file management, lack of information access, unfair distribution of goods, and management difficulties in Consumer Cooperative Stores. In Phase Two, an appropriate technology and action plan was developed to address the selected problem by designing a digital cooperative store management system.

In this intervention phase, the group implemented a fully working web-based system using React for the frontend and Laravel for the backend. The website was also prepared to work as a Telegram Mini App so that cooperative members and store users can access the service through Telegram without installing a separate mobile application. The implemented system includes member registration, login, product listing, inventory management, order placement, order history, admin dashboard, identity verification, reports, notifications, and Telegram access.

The system was tested by store managers, cooperative members, and community representatives. The intervention reduced dependence on manual processes by providing digital records, easier inventory tracking, transparent product information, controlled ordering, and improved communication between the store and members. The major challenges observed during implementation were limited user familiarity with digital systems and incomplete store data. Based on the results, the group recommends expanding the system to other cooperative stores after additional training, data preparation, and continuous technical support.

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1. PROJECT OVERVIEW

Community Based Training Programme is an important practical learning approach that enables students to identify community problems, design appropriate solutions, and apply their knowledge through real intervention. The project was conducted in Bosa Addis Kebele, Jimma City, with a focus on the Consumer Cooperative Stores operating in the area.

During Phase One, the group carried out a community survey and identified several technology-related problems. These included lack of technological awareness, limited information access, insufficient technology resources, and dependence on manual systems. After further discussion and prioritization, the group selected the problem of manual and undigitized cooperative store management because it directly affected the fairness, transparency, and efficiency of goods distribution.

During Phase Two, the group developed an appropriate technology and action plan. The proposed solution was a digital cooperative store management system designed to support member registration, inventory management, product information, order placement, identity verification, reporting, and communication. The design included system requirements, entities, relationships, database normalization, technology stack, testing plan, deployment plan, and user training plan.

In Phase Three, the group implemented the planned solution as a fully working system. The system was developed as a website using React and Laravel and was prepared to operate as a Telegram Mini App through Telegram Web Apps. This approach was selected because many users are already familiar with Telegram, and Telegram Mini Apps allow users to access a web application directly inside Telegram without installing a separate mobile application.

The intervention focused on converting the major manual activities of the Consumer Cooperative Store into a digital workflow. Store managers can manage products and inventory, members can register and view products, orders can be placed digitally, and administrators can monitor transactions and reports. The system was tested with store managers, cooperative members, and community representatives to evaluate usability, usefulness, and readiness for wider use.

1.1 Background of the Selected Problem

Consumer Cooperative Stores are important for supplying essential goods to community members. However, when the store management process depends mainly on manual records, several problems can occur. Manual registration and paper-based order handling can lead to delays, misplaced records, unclear inventory information, and difficulty in confirming whether goods are distributed fairly.

The Phase One and Phase Two studies showed that cooperative members needed better access to information about available products, prices, and distribution activities. Store managers also needed a simpler way to manage stock levels, member information, and transaction records. Therefore, the digital system was implemented to reduce manual work and improve transparency.

1.2 Summary of Previous Phases

Phase	Main Activity	Output
Phase One	Community problem identification and demographic survey	Identification of technology-related problems and selection of cooperative store management as the focus area
Phase Two	Appropriate technology and action plan development	Design of digital cooperative store system, database structure, features, action plan, and deployment approach
Phase Three	Intervention and implementation	Working React and Laravel website integrated as Telegram Mini App, tested by store managers, members, and representatives

2. OBJECTIVES AND SCOPE

2.1 General Objective

The general objective of this phase was to implement and test a digital cooperative store management system that reduces manual processes and improves transparency, accessibility, and efficiency in Consumer Cooperative Stores in Bosa Addis Kebele.

2.2 Specific Objectives

- To implement a working website using React and Laravel based on the system design prepared in Phase Two.
- To prepare the website for access through Telegram Mini App so that users can open the system from Telegram.
- To provide member registration, login, product listing, inventory management, order placement, order history, admin dashboard, identity verification, reports, notifications, and Telegram access.
- To test the system with store managers, cooperative members, and community representatives.
- To reduce dependence on manual records by introducing digital registration, inventory, order, and reporting workflows.
- To collect user feedback and identify challenges that need improvement before wider deployment.

2.3 Scope of the Intervention

The intervention was limited to Consumer Cooperative Stores found in Bosa Addis Kebele. The implemented system supports the core store activities needed by cooperative members, store managers, and administrators. The scope includes member management, product and inventory management, order management, identity verification, reporting, notification, and Telegram-based access.

The intervention did not include full integration with external payment systems, national ID systems, or large-scale government databases. These areas are recommended for future expansion after the system is accepted and stabilized at the cooperative store level.

2.4 Target Users

User Group	Role in the System
Cooperative Members	Register, login, view products, place orders, view order history, receive notifications, and access information through Telegram
Store Managers	Manage products, update inventory, review orders, verify member activity, and reduce manual record keeping
Administrative Staff	Manage users, verify identity, generate reports, and monitor store operations
Community Representatives	Observe the system, provide feedback, and support adoption in the community

3. INTERVENTION DETAILS

The intervention was carried out by implementing the system that was designed during Phase Two. The implementation followed a practical approach: development of the web system, preparation of Telegram Mini App access, insertion of store data, testing of each feature, demonstration to users, collection of feedback, and preparation of recommendations for expansion.

3.1 Development Environment and Technology Used

Component	Technology Used	Purpose
Frontend	React	To build the user interface for members, store managers, and administrators
Backend	Laravel	To manage APIs, authentication, business logic, database interaction, and reports
Database	Relational database connected to Laravel	To store members, products, orders, inventory, verification records, reports, and messages
Access Platform	Website and Telegram Mini App	To allow users to access the system through browser and Telegram
Testing Method	Manual functional testing and user testing	To check whether all major features work correctly and are understandable to users

3.2 System Architecture

The implemented system follows a client-server architecture. The React frontend provides the visible screens and interaction forms. The Laravel backend handles authentication, validation,

database operations, and communication between users and stored data. The Telegram Mini App layer allows the same web system to open inside Telegram using Telegram Web Apps.

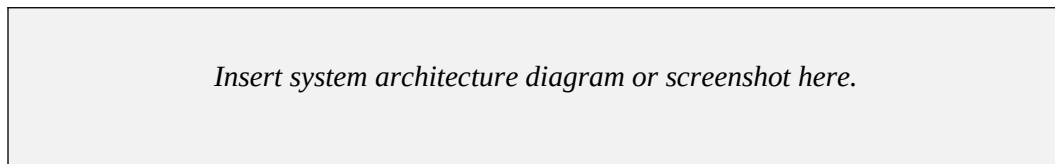


Figure 3.1: Implemented System Architecture

3.3 Features Implemented

No.	Feature	Implementation Status	Purpose
1	Member Registration	Completed	Allows cooperative members to create digital profiles
2	Login and Authentication	Completed	Protects user accounts and separates member and admin access
3	Product Listing	Completed	Displays available products, prices, and descriptions
4	Product Inventory Management	Completed	Allows managers to update product quantity and stock information
5	Order Placement	Completed	Allows members to request products digitally
6	Order History	Completed	Allows members and admins to review previous orders
7	Admin Dashboard	Completed	Allows managers to

			monitor users, products, orders, and system activity
8	Identity Verification	Completed	Supports verification of cooperative members before service delivery
9	Reports	Completed	Generates useful summaries for inventory, orders, and member activities
10	Notifications	Completed	Informs users about order status and important updates
11	Telegram Access	Completed	Allows the website to work as a Telegram Mini App

3.4 Member Registration and Authentication

The system allows cooperative members to register by providing their basic information. After registration, users can login securely and access the member side of the system. Authentication helps to ensure that only registered users can place orders, view their own order history, and receive account-related updates.

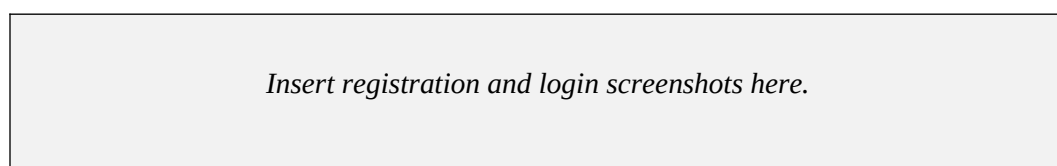


Figure 3.2: Member Registration and Login Screens

3.5 Product Listing and Inventory Management

The product listing feature enables members to view available goods, prices, descriptions, and availability. The inventory management feature allows store managers to add products, update stock quantities, edit prices, and remove unavailable products. This reduces dependence on handwritten stock records and improves the accuracy of product information.

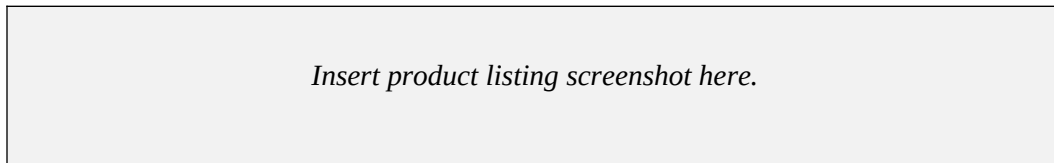


Figure 3.3: Product Listing Screen

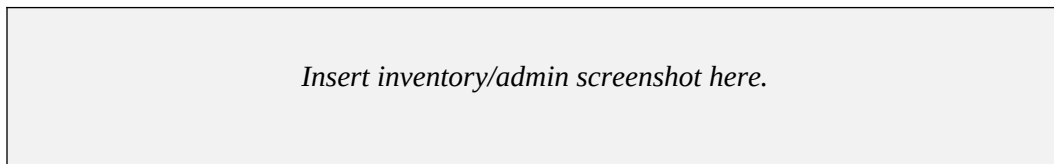


Figure 3.4: Admin Inventory Management Screen

3.6 Order Placement and Order History

The order placement feature allows members to select products and submit orders digitally. After submitting an order, the system stores the order record and makes it available for review by the store manager. The order history feature allows members and administrators to track past orders, reducing disagreement and confusion about previous transactions.

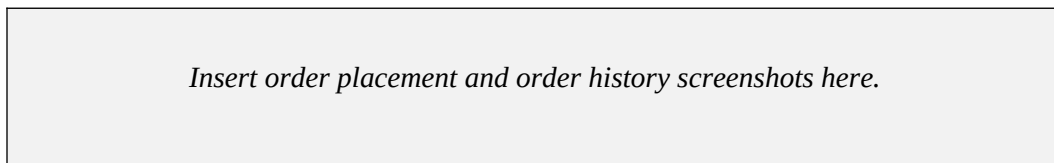


Figure 3.5: Order Placement and Order History Screens

3.7 Admin Dashboard and Reports

The admin dashboard gives store managers and administrative staff a central place to view and manage the system. The dashboard includes information about members, products, inventory,

orders, identity verification, reports, and notifications. Reports help the store understand stock movement, order volume, member activity, and system usage.

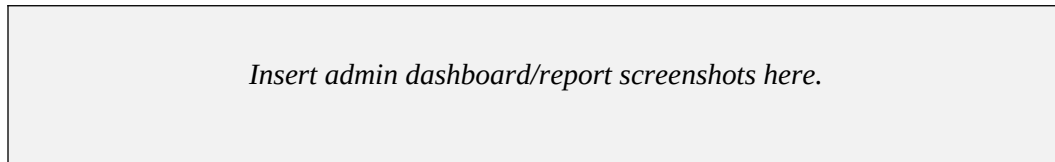


Figure 3.6: Admin Dashboard and Report Screen

3.8 Identity Verification

Identity verification was implemented to reduce errors and fraudulent activity during service delivery. Through the system, member information can be recorded and verified before the member uses cooperative store services. This helps the store confirm that goods are provided to the correct members and supports fair distribution.

3.9 Telegram Mini App Access

The website was prepared to work as a Telegram Mini App using Telegram Web Apps. This allows the application to open inside Telegram, which is useful because many smartphone users are already familiar with Telegram. This approach reduces the need for users to download and install a separate application. It also supports easier notification and communication with cooperative members.

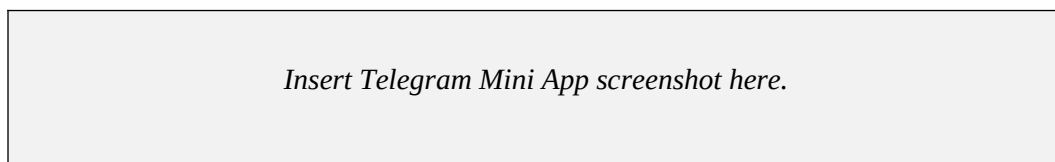


Figure 3.7: Telegram Mini App Access Screen

3.10 Testing and Demonstration

After development, the group tested the implemented features and demonstrated the system to the intended users. Store managers, cooperative members, and community representatives participated in testing and observation. The testing focused on whether users could understand

the system, whether the main features worked correctly, and whether the system could reduce manual work in daily cooperative store activities.

Test Area	Test Activity	Result
Registration	Create member account using sample and real member information	Successful
Login	Login using registered user credentials	Successful
Product Listing	View available products and prices	Successful
Inventory Management	Add, edit, and update product stock	Successful
Order Placement	Submit order as a member	Successful
Order History	Review previous orders	Successful
Admin Dashboard	View and manage system records	Successful
Identity Verification	Verify member information	Successful
Reports	Generate summary reports	Successful
Notifications	Send and view system notifications	Successful
Telegram Access	Open the system as a Telegram Mini App	Successful

Insert photos of system demonstration with users here.

Figure 3.8: User Testing and Demonstration Evidence

3.11 Training and User Support

Because some users were not familiar with digital systems, the group provided simple user guidance during the demonstration. The training focused on how to open the system, register, login, view products, place orders, and check order status. Store managers were also shown how to add products, update inventory, verify members, and view reports.

Training Topic	Target User	Purpose
How to access the website and Telegram Mini App	Members and managers	To make the system easy to open and use
Member registration and login	Cooperative members	To help users create and access their accounts
Product browsing and order placement	Cooperative members	To enable digital ordering
Inventory and order management	Store managers	To reduce manual stock and order handling
Reports and verification	Administrative staff	To support transparent management and correct service delivery

3.12 Implementation Challenges

The implementation was successful, but the group faced some practical challenges during the intervention.

- Some users were not familiar with digital systems. This required additional explanation and hands-on demonstration.
- Store data was incomplete. Some product and member records had to be organized before they could be entered into the system.
- Different users had different levels of digital literacy, so the interface and training needed to be simple and direct.

4. SIGNIFICANCE AND IMPACT

The intervention created practical value for the Consumer Cooperative Store and the surrounding community. The system showed how a manual cooperative store process can be transformed into a digital workflow that is more transparent, organized, and easier to manage.

4.1 Before and After Comparison

Before the Intervention	After the Intervention
Member and order information was mainly handled manually.	Member and order information can be recorded digitally.
Store managers depended on paper-based records and memory.	Store managers can use the admin dashboard and reports.
Product availability and prices were not easily accessible to all members.	Members can view products, prices, and availability through the system.
Order history was difficult to track clearly.	Order history is stored and can be reviewed.
Identity verification was slower and more error-prone.	Identity verification can be managed digitally.
Communication with members was limited.	Notifications can be used to share updates.

4.2 Reduction of Manual Process

The main result of the intervention was the reduction of manual processes. The digital system reduces the need to repeatedly write member information, product lists, order details, and inventory records by hand. It also reduces the chance of losing records because information is stored in the system and can be retrieved when needed.

4.3 Improvement in Transparency

The system improves transparency by making product information, order records, inventory status, and reports easier to access. Cooperative members can view available products and prices, while store managers can track orders and stock movement. This supports fairer distribution because the system keeps records that can be checked later.

4.4 Improvement in Store Management

The admin dashboard and reporting features help store managers perform their tasks more efficiently. Instead of depending only on paper records, managers can update products, monitor inventory, review orders, verify members, and generate reports from one system. This reduces workload and makes decision-making easier.

4.5 Community Response and Feedback

Store managers, cooperative members, and community representatives tested the system and observed its usefulness. The feedback showed that users appreciated the ability to view products digitally, place orders, and reduce manual recording. However, the testing also showed that continuous user training is necessary because some users are not yet comfortable with digital systems.

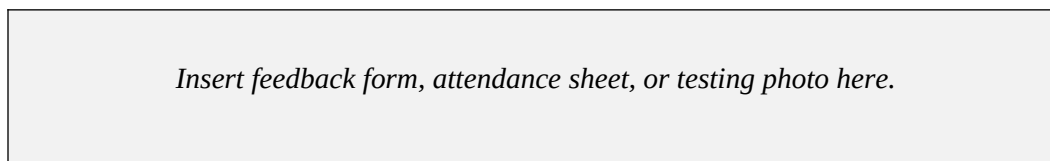


Figure 4.1: Community Feedback and Testing Evidence

4.6 Sustainability of the Intervention

The intervention can be sustained if the cooperative store assigns responsible users to manage product data, update inventory, verify members, and support users. Since the system is web-based and accessible through Telegram, it can be maintained and expanded without requiring every user to install a separate application. Continued training and data quality improvement are important for long-term success.

5. RECOMMENDATIONS

Based on the implementation, testing, and feedback from users, the group forwards the following recommendations.

5.1 Expansion to Other Cooperative Stores

The system should be expanded to other cooperative stores after further testing and improvement. The problem of manual records, unclear inventory information, and inefficient order management is not limited to one store. Therefore, the solution can support other cooperative stores that face similar challenges.

5.2 Continuous User Training

Because some users are not familiar with digital systems, continuous training should be provided. Training should focus on basic smartphone use, Telegram access, member registration, order placement, checking order status, and understanding notifications. Store managers should receive additional training on inventory updates, reports, and user management.

5.3 Data Preparation and Updating

Before expanding the system, cooperative stores should organize member records, product lists, prices, and existing inventory data. Clean and complete data is necessary for the system to work accurately. A responsible person should be assigned to update data regularly.

5.4 Technical Support and Maintenance

The system should have technical support for fixing bugs, improving features, managing hosting, and supporting users. Future student groups or responsible technical bodies can continue improving the system based on user feedback.

5.5 Future Feature Improvements

- Integrate digital payment options such as Telebirr when the cooperative store is ready.
- Add stronger analytics for monthly sales, stock movement, and member activity.
- Add multi-store support so that one system can manage several cooperative stores.
- Add offline data entry support for times when internet connectivity is weak.
- Improve notification and announcement features for community-wide updates.

5.6 Conclusion

The Phase Three intervention successfully transformed the planned solution into a fully working system. By implementing a React and Laravel website prepared for Telegram Mini App access, the group reduced manual processes in Consumer Cooperative Store activities and demonstrated the practical value of digital technology in Bosa Addis Kebele. The system supports registration, login, product listing, inventory management, order placement, order history, admin dashboard, identity verification, reports, notifications, and Telegram access. The intervention showed positive impact, especially in reducing manual records and improving access to information. With continuous training, better data preparation, and expansion to other cooperative stores, the system can become a sustainable solution for improving cooperative store management.

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Community Based Training Programme in Bosa Addis Kebele, Report of Phase I, Computer Science Section 2, Group 5 Students, April 2024.

Appropriate Technology and Action Plan Development in Bosa Addis Kebele, Report on Phase II, Computer Science Section 2, Group 5 Students, June 2024.

Telegram Bot Platform Documentation, Telegram Web Apps for Bots,
<https://core.telegram.org/bots/webapps>

APPENDIX

Appendix A: Evidence to Attach Before Final Submission

- Screenshot of member registration page.
- Screenshot of login page.
- Screenshot of product listing page.
- Screenshot of order placement page and order history page.
- Screenshot of admin dashboard.
- Screenshot of inventory management page.
- Screenshot of report page.
- Screenshot of Telegram Mini App access.
- Photos of group members demonstrating the system to store managers and cooperative members.
- Photos of the previous manual process or paper records, if available.
- User feedback forms or short written comments from testers.

Appendix B: Sample User Feedback Summary

User Type	Feedback Summary
Store Manager	The system makes product and order management easier than manual records.
Cooperative Member	The system helps members view products and place orders more clearly.
Community Representative	The system can improve transparency if users receive enough training.

Appendix C: Final Deliverables

- Phase Three Intervention Report in hard copy and soft copy.
- Working React and Laravel website.
- Telegram Mini App access configuration.
- Screenshots and photos showing implementation and testing.
- Source code and technical documentation.
- User guide for members and store managers.